ModelIt!

Student Activity Pack

The Coral Reef Death Spiral

Modeling Thermal Bleaching, Recovery Thresholds, and Climate Feedback

Grade 11L2

NGSS HS-LS2-2, HS-LS2-6

Stability and Change / Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think causes coral to turn white, and can it recover?**

2. **Why might rising ocean temperatures by just 1-2 degrees be dangerous for marine life?**

3. **Draw a diagram showing the relationship between a coral animal and the tiny organisms that live inside it.**

4. **If 25% of marine species depend on coral reefs, what would happen to the ocean if reefs disappeared?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sea Surface Temperature | Bleaching Frequency | Coral Recovery Rate | Reef Structural Complexity | Marine Biodiversity | Algae Competition

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Historical Baseline

Bleaching every 25-30 years, full recovery between events

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Current Trajectory

Bleaching every 6 years, incomplete recovery

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Accelerated Warming

Annual bleaching, no recovery window

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Ocean Acidification Interaction — Lower pH compounds thermal stress by making it harder for corals to rebuild calcium carbonate skeletons between bleaching events, effectively lengthening recovery time

Herbivore Fish Population — Parrotfish and surgeonfish that graze algae off reef surfaces, preventing algae overgrowth and keeping space open for coral recolonization — overfishing removes this critical reef maintenance function

Coral Genetic Diversity — The range of heat-tolerance genes within a coral population, which determines the reef's ability to adapt to warming — low diversity means fewer individuals can survive higher temperatures

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How much heat can a coral reef absorb before it crosses the point of no return — and what happens to the 25% of marine life that depends on it? Use evidence from your simulation to support your answers.

1. **Why is the ratio of bleaching frequency to recovery time more important than the severity of any single bleaching event?**

2. **How does your model demonstrate that a phase shift from coral to algae is a tipping point rather than a gradual transition?**

3. **What feedback loops in your model make reef degradation self-reinforcing once it begins?**

4. **How would reducing local stressors (pollution, overfishing) change the recovery dynamics in your model even without reducing ocean temperature?**

5. **What does your model suggest about the maximum warming the world's coral reefs can survive?**

8 STEM Challenge: Design a Coral Reef Resilience Strategy

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YOUR ENGINEERING MISSION

Design an evidence-based intervention plan that maximizes a coral reef's chances of surviving the next 30 years of ocean warming.

THE SCENARIO

A marine protected area with a degraded but still living coral reef has secured funding for a 10-year resilience project. Your team must design a strategy that addresses both local stressors (water quality, overfishing, algae competition) and the global thermal stress that drives bleaching — knowing that you cannot control ocean temperature directly.

Engineering Design Thinking

1. HOW WOULD YOU MAP THE REEF TO IDENTIFY THE MOST RESILIENT CORAL POPULATIONS AND PRIORITIZE PROTECTION ZONES?
2. WHAT WATER QUALITY INTERVENTIONS (SEDIMENT REDUCTION, NUTRIENT MANAGEMENT) WOULD GIVE CORALS THE BEST RECOVERY CONDITIONS?
3. HOW WOULD YOU DESIGN A CORAL NURSERY PROGRAM TO PROPAGATE HEAT-TOLERANT GENOTYPES FOR REEF RESTORATION?
4. HOW WOULD YOU MEASURE SUCCESS — WHAT METRICS WOULD YOU TRACK AND AT WHAT INTERVALS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Marine Ecologists and Coral Reef Scientists study reef ecosystems, assess bleaching damage, and design restoration strategies. They work for organizations like the Great Barrier Reef Marine Park Authority, NOAA Coral Reef Conservation Program, and research universities, earning $65,000–$130,000/year. Coral Restoration Technicians specializing in nursery propagation and outplanting earn $45,000–$75,000/year.